

Maharashtra State Board Of Technical Education, Mumbai																									
Learning and Assessment Scheme Of Diploma Courses for Working Professionals																									
Programme Name		: Diploma In Civil Engineering (Working Professional)																							
Programme Code		: PCE										With Effect From Academic Year					: 2024-25								
Duration of Programme		: 6 Semester																							
Semester		: Third										Scheme					: F								
Sr No	Course Title	Abbreviation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme												
							Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning	Total Marks			
							CL	TL	LL								Practical								
														FA-TH	SA-TH	Total		FA-PR		SA-PR			SLA		
																Max	Min	Max	Min	Max	Min		Max	Min	
(All Compulsory)																									
1	ADVANCED SURVEYING	ASU	SEC	313321	No	0	3		4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175	
2	CONCRETE TECHNOLOGY	CTE	DSC	313322	No	1	3		2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175	
3	HIGHWAY ENGINEERING	HEN	DSC	313323	No	1	3		2	1	6	3	3	30	70	100	40	25	10	-	-	25	10	150	
4	CONSTRUCTION MANAGEMENT	CMA	DSC	313010	No	0	1		2	1	4	2	-	-	-	-	25	10	-	-	25	10	50		
Total						2	10	0	10	4		12		90	210	300		100		50		100		550	
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. 7. * Self learning includes micro project / assignment / other activities. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)																									

ADVANCED SURVEYING**Course Code : 313321**

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Third
Course Title	: ADVANCED SURVEYING
Course Code	: 313321

I. RATIONALE

The technology has brought the significant advancements in field of surveying. This will help civil engineers for accurate measurements of physical features of various construction projects and land with utmost accuracy, speed and easy operation of these surveying equipment's. The data obtained by various advanced surveying equipment's includes information of topography, grading, elevation, distances etc. Such data obtained helps civil engineers for future project planning and effective execution. The advanced surveying also helps in identifying potential risks associated with construction projects. This course will help students to acquire skills associated with surveying of land and buildings.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Use the relevant instrument to undertake the survey of the given area.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use the Tacheometer to obtain relevant details of the terrain in given situation.
- CO2 - Set out a Simple Circular curve to finalize the alignment of the given element.
- CO3 - Prepare layout plans using relevant surveying instruments.
- CO4 - Locate the co-ordinates of a given stations using the relevant technology.

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- CO5 - Interpret the images of given terrain using Photogrammetry Techniques.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313321	ADVANCED SURVEYING	ASU	SEC	3	-	4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

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ADVANCED SURVEYING**Course Code : 313321****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the principles of Tacheometric surveying. TLO 1.2 Use the tacheometer to determine Reduced level, horizontal and vertical distance in the given situation. TLO 1.3 Select the relevant method of Tacheometric surveying in the given situation. TLO 1.4 Calculate constants of a Tacheometer from the given data. TLO 1.5 Specify the Limitations of tacheometry with examples.	Unit - I Tacheometric Surveying 1.1 Principle of tacheometry, Use of Tacheometry 1.2 Tacheometer and its component parts, Analytic lens, Tacheometric formula for horizontal distance with telescope horizontal and staff vertical. 1.3 Methods of Tacheometry: Stadia and fixed hair method 1.4 Field method for determining constants of tacheometer 1.5 Limitations of tacheometry.	Video Demonstrations Presentations Collaborative learning Hands-on Model Demonstration Lecture Using Chalk-Board Demonstration

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Classify the curves used in surveying to fix the alignment. TLO 2.2 Draw the labelled diagram of simple circular curve. TLO 2.3 Derive the relationship between Radius and Degree of curve TLO 2.4 Set a simple circular curve using the relevant method of curve setting in the given situation.	Unit - II Curves setting 2.1 Curve: Definition, Necessity of Curves, Types of curves used in roads and railway alignments. 2.2 Elements of simple circular curve, Designation of the curve by Radius and Degree of curve. 2.3 Radius and Degree of curve. 2.4 Setting out a simple circular curve by offsets from long chord and Rankine's method of deflection angles.	Presentations Lecture Using Chalk-Board Collaborative learning Video Demonstrations Demonstration Model Demonstration Hands-on

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Use the EDM to measure the distance between two given stations. TLO 3.2 Use Electronic Digital Theodolite to measure the required angle. TLO 3.3 Explain the procedure to measure the angle between the given lines using Total Station instrument. TLO 3.4 Undertake the site layout operation for the given building structure using Total Station instrument.	Unit - III Advanced Surveying Equipment's 3.1 Electronic Distance Meter (EDM): Principle of Electronic Distance Meter (EDM), component parts and their Functions, use of EDM. 3.2 Electronic Digital Theodolite: Construction and Features of Electronic Digital Theodolite, procedure of angle measurement. 3.3 Total Station: Introduction, component parts with their functions, and Applications of Total Station, Temporary adjustments, sources of errors in Total Station, Measurements of Horizontal angles, vertical angles, distances and coordinates using Total Station. Traversing, Profile Survey and Contouring with Total Station. 3.4 Building Site layout using Total Station: Procedure.	Model Demonstration Video Demonstrations Hands-on Presentations Lecture Using Chalk-Board Collaborative learning Demonstration Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Propose the relevant system of remote sensing to be used for the given situation. TLO 4.2 Describe the procedure to find out the coordinates of the given station using GPS. TLO 4.3 Use the GIS technology to solve the given civil engineering problem.	Unit - IV Remote sensing, GPS and GIS 4.1 Remote Sensing : Definition, Electro-Magnetic Energy, Active and Passive system, Applications of remote sensing in Mining, land use / Land cover, mapping, disaster management and Environment. 4.2 Global Positioning System: Introduction, Construction and use of Global Positioning System (G.P.S.) 4.3 Geographic Information System (GIS): Overview, Component, Sources of errors, applications, Software's in GIS.	Presentations Video Demonstrations Collaborative learning Lecture Using Chalk-Board Demonstration Site/Industry Visit Case Study Hands-on

ADVANCED SURVEYING**Course Code : 313321**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Suggest the relevant method of aerial surveying for the given situation. TLO 5.2 Classify the type of drones as per DGCA TLO 5.3 Explain the principles of photogrammetry TLO 5.4 Explain the type of Photogrammetry. TLO 5.5 Specify the Merits and Demerits of Photogrammetry surveying. TLO 5.6 Use the photogrammetry techniques to solve civil engineering problems.	Unit - V Aerial Surveying and Photogrammetry 5.1 Aerial surveying: Definition, principle, uses, methods 5.2 DGCA Classification of Drones, Silent features of Drone Rules, 2021 as per DGCA. 5.3 Definition of photogrammetry, Basic Principles of Photogrammetry. Types of Photogrammetry: Terrestrial and Aerial Photogrammetry 5.4 Types of Photographs, Terminology in aerial surveying. 5.5 Merits and Demerits of Photogrammetry surveying 5.6 Applications of Photogrammetry in civil engineering.	Case Study Video Demonstrations Demonstration Hands-on Collaborative learning Lecture Using Chalk-Board Presentations Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Use theodolite as a Tacheometer to determine the Tacheometric Constant	1	*Determine the Tacheometric Constant	2	CO1

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 2.1 Use theodolite as a Tacheometer to compute reduced levels and horizontal distances	2	*Determine reduced levels of given stations and horizontal distances by Tacheometric Method (Part I)	2	CO1
LLO 3.1 Use theodolite as a Tacheometer to compute reduced levels and horizontal distances	3	*Determine reduced levels of given stations and horizontal distances by Tacheometric Method (Part II)	2	CO1
LLO 4.1 Use the offsets from Long Chord Method to Set out a circular curve	4	*Setting out of a circular curve by offsets from Long Chord Method.	2	CO2
LLO 5.1 Apply the technique of Rankine's Method of Deflection Angles to Set out a circular curve	5	*Setting out a circular curve by Rankine's Method of Deflection Angles. (Project) (Part I). Plot the curve details on A1 size imperial drawing sheet.	2	CO2
LLO 6.1 Apply the technique of Rankine's Method of Deflection Angles to Set out a circular curve	6	*Setting out a circular curve by Rankine's Method of Deflection Angles. (Project) (Part II). Plot the curve details on A1 size imperial drawing sheet.	2	CO2
LLO 7.1 Use EDM to measure horizontal and vertical distance	7	*Determine horizontal and vertical distance by using EDM.	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 8.1 Use Electronic Digital Theodolite to determine Horizontal and vertical angles	8	*Determine Horizontal and vertical angles using Electronic Digital Theodolite	2	CO3
LLO 9.1 Set up the Total Station instrument.	9	Setting up the Total Station instrument on site for surveying.	2	CO3
LLO 10.1 Use Total station instrument to measure horizontal, vertical and slope distances	10	*Determine horizontal, vertical and slope distances using Total station equipment (Part I)	2	CO3
LLO 11.1 Use Total station instrument to measure horizontal, vertical and slope distances	11	*Determine horizontal, vertical and slope distances using Total station equipment. (Part II)	2	CO3
LLO 12.1 Use Total station instrument to measure the given horizontal and vertical angles	12	*Determine horizontal and vertical angles using Total Station. (Part I)	2	CO3
LLO 13.1 Use Total station instrument to measure the given horizontal and vertical angles	13	*Determine horizontal and vertical angles using Total Station. (Part II)	2	CO3
LLO 14.1 Use Total station to determine Reduce Levels	14	*Determine the Reduced Levels of given stations (Minimum 10 station) (Part I)	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 15.1 Use Total station to determine Reduce Levels	15	*Determine the Reduced Levels of given stations (Minimum 10 station) (Part II)	2	CO3
LLO 16.1 Use Total Station to stack out station points on ground.	16	Stack out (transferring the data on ground) using Total Station (Part I)	2	CO3
LLO 17.1 Use Total Station to stack out station points on ground.	17	Stack out (transferring the data on ground) using Total Station (Part II)	2	CO3
LLO 18.1 Use Total station instrument to measure Reduced Level for given road profile project	18	Road profile of 100m length using Total Station instrument (Part I)	2	CO3
LLO 19.1 Use Total station instrument to measure Reduced Level for given road profile project	19	Road profile of 100m length using Total Station instrument (Part II)	2	CO3
LLO 20.1 Use Total station instrument to prepare contour plans	20	Contouring using Total Station instruments for the area of size 50 X 50 m	2	CO3
LLO 21.1 Use Total Station to prepare Building site layout	21	*Prepare Building site layout by using Total Station (Project) (Part I). Plot the project details on A1 size imperial drawing sheet.	2	CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 22.1 Use Total Station to prepare Building site layout	22	*Prepare Building site layout by using Total Station (Project) (Part II). Plot the project details on A1 size imperial drawing sheet.	2	CO3
LLO 23.1 Use Total station to carry out five-sided closed traverse Surveying Project	23	*Carry out 5-Sided closed traverse Surveying project by using Total Station. (Project) Part I). Plot the traverse details on A1 size imperial drawing sheet.	2	CO3
LLO 24.1 Use Total station to carry out five-sided closed traverse Surveying Project	24	*Carry out 5-Sided closed traverse Surveying project by using Total Station. (Project) Part II). Plot the traverse details on A1 size imperial drawing sheet.	2	CO3
LLO 25.1 Use GPS technology to locate the coordinates of a station.	25	*Locate the coordinates of a station with the help of GPS.	2	CO4
LLO 26.1 Develop the contour maps using photogrammetry images	26	Create the images of contouring map with given data (Photogrammetry images, etc) using the freeware/open source software (Part I)	2	CO5
LLO 27.1 Develop the contour maps using photogrammetry images	27	Create the images of contouring map with given data (Photogrammetry images, etc) using the freeware/open source software (Part II)	2	CO5
LLO 28.1 Develop the Road Profile plan using photogrammetry images	28	Create the images of Road Profile plan with given data (Photogrammetry images, etc) using the freeware/open source software (Part I)	2	CO5
LLO 29.1 Develop the Road Profile plan using photogrammetry images	29	Create the images of Road Profile plan with given data (Photogrammetry images, etc) using the freeware/open source software (Part II)	2	CO5

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 30.1 Use relevant software for preparation of contour maps using given image data	30	*Write a brief report on the visit to nearby surveying software laboratory for visualization of image creation of contouring map of given area using data collected by drone technology / Arrange Expert Lecture / Show study videos of Photogrammetry surveying.	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Assignment

- Measure the height of the flag post in your institute using Theodolite as tacheometer.
- Measure the height of the slab bottom of second floor of your institute building using Theodolite as tacheometer.
- Set the alignment of proposed road using Theodolite as tacheometer.
- Use freeware or open source software for image processing using photogrammetry principles
- Plot the contours using Total station by direct method.
- Mark building layout using Total station.
- Measure distance between two distant(>500m) points using EDM
- Locate the coordinates of the campus using GPS

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- Search and download the freeware/open source software and prepare a report stating the applications.

Micro project

- Carry out comparative study of following survey instruments of different make and brands: Total station/ EDM/GPS/Digital theodolite.
 - Collect the relevant technical and commercial information of minimum five advanced survey instruments available in the market with specifications.
 - Determine the RLs of the existing structures like lintels, chajja, slab, and beam using Tacheometer and Total station in a multistoried building and compare the results.
 - Download specifications for Total Station/ EDM/GPS and make a chart.
 - Set the profiles of curves at the changes in alignment of road in the premises of the institute (minimum two).
 - Study the specifications of Mobile devices used for distance measurement.
 - Collect the information on 360-degree laser
 - Collect information of software required for mapping of images for photogrammetry.
 - Collect the Information about Drone survey.
 - Collect the information on Rover survey for land measurement
-

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- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Twenty Second Transit theodolite with accessories.	1,2,3,4,5,6
2	GPS instrument	25
3	Surveying Drone - About 1 to 1.5 Sq. Km. area can be easily captured by one PPK Survey grade Drone flying with an altitude of about 80 to 100m above average ground level)	27,26,28,29,30
4	Electronic Distance meter (+or- 2mm accuracy) with accessories.	7
5	Electronic Digital Theodolite with accessories.	8
6	Total Station (+ or - 2mm accuracy) instrument with accessories	9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)**MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

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Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Tacheometric Surveying	CO1	10	2	8	6	16
2	II	Curves setting	CO2	8	2	4	6	12
3	III	Advanced Surveying Equipment's	CO3	10	2	8	6	16
4	IV	Remote sensing, GPS and GIS	CO4	9	4	4	6	14
5	V	Aerial Surveying and Photogrammetry	CO5	8	4	4	4	12
Grand Total				45	14	28	28	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Termwork, Assignment, Microproject (60% Weightage to process and 40% weightage to product), Question and Answer

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test), Practical Exam

XI. SUGGESTED COS - POS MATRIX FORM

ADVANCED SURVEYING**Course Code : 313321**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	3	1	3	2	1	2			
CO2	3	3	2	2	2	1	2			
CO3	3	3	3	3	2	2	3			
CO4	3	2	2	2	2	1	3			
CO5	3	1	1	2	2	2	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Kanetkar T. P.; Kulkarni S. V.	Surveying and Levelling volume I & II	Pune Vidyarthi Gruh Prakashan, Pune; ISBN:978-81-858-2511-3
2	Basak N. N.	Surveying and Levelling	McGraw Hill Education, New Delhi ISBN 93-3290-153-8

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Sr.No	Author	Title	Publisher with ISBN Number
3	S. K. Duggal	Surveying I & II	McGraw Hill Education, New Delhi, ISBN: 978-00-701-5137-6
4	Punmia B.C, Ashok Kumar Jain, Arun Kumar	Surveying I & II	Laxmi Publications., New Delhi. ISBN: 8- 17-008853-4
5	Shivam Pandey	Basic Concept of Remote Sensing, GPS, and GIS	Sankalp Publication, Gaurav Path, Bilaspur Chhathisgarh-4955001 ISBN: 978-81-94-77801-1

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://archive.nptel.ac.in/content/storage2/courses/105107122/modules/module7/html/100.htm	Tacheometry Surveying
2	https://www.youtube.com/watch?v=7UhaCqea7IY	Curve Setting
3	https://archive.nptel.ac.in/content/storage2/courses/105107122/modules/module11/index.htm	Curve Setting
4	https://nptel.ac.in/courses/105104100	Lecture on Total Station
5	https://www.youtube.com/watch?v=bbs5AEPstl4	Total Station
6	https://www.youtube.com/watch?v=1KCqxx8r5Y4	Electronic Digital Theodolite
7	https://www.youtube.com/watch?v=QLgwwVdMaWU	Remote sensing GIS and GPS
8	https://archive.nptel.ac.in/courses/105/103/105103193/	Remote Sensing and GIS
9	https://onlinecourses.nptel.ac.in/noc22_ce84/preview	Remote Sensing and GIS
10	https://archive.nptel.ac.in/courses/105/104/105104101/	Aerial Surveying and Photogrammetry
11	https://nptel.ac.in/courses/105104100	Aerial Surveying and Photogrammetry

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Sr.No	Link / Portal	Description
Note :		
Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

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CONCRETE TECHNOLOGY

Course Code : 313322

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Third
Course Title	: CONCRETE TECHNOLOGY
Course Code	: 313322

I. RATIONALE

Concrete is the most common construction material on earth and it directly determines the construction quality of the concrete structure and plays a very important role in the stability of the building structure. A diploma civil engineering students are required to develop the basic competency in dealing with concrete as a construction material ,so that they should have the basic knowledge of cement chemistry, concrete batching, placement, compaction and curing and testing of concrete. This course will be helpful for students to build their theoretical and practical knowledge in civil and structural engineering.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Design the concrete mix as per the given specifications.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Suggest relevant types of cement to be used in the given site conditions.
- CO2 - Classify the given aggregates based on its shape and size with the importance of their properties.
- CO3 - Prepare concrete of required specifications in the given situation.
- CO4 - Undertake the necessary procedures to maintain the quality of given type of concrete.

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- CO5 - Suggest relevant type of admixtures to be used in the given situation.

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Total IKS Hrs for Sem. : 1 Hrs

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Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

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CONCRETE TECHNOLOGY**Course Code : 313322****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the chemical constituents of the given type of cement TLO 1.2 Describe the relevant physical properties of the given type of cement in the given situation. TLO 1.3 Carry out the relevant test on the given sample of cement for the given purpose. TLO 1.4 Propose the relevant type of cement to be used in the given situation.	Unit - I Cement 1.1 Chemical Constituents of OPC (Ordinary Portland Cement) and their effects on properties of OPC, Bogue's compounds and their properties, hydration of cement. Physical properties of OPC: fineness, specific gravity, standard consistency, setting time, soundness, compressive strength 1.2 Different grades of OPC 33, 43, and 53 with its uses and specifications of physical properties as per relevant IS codes for OPC. 1.3 Testing of OPC: Field tests and laboratory tests-fineness test, standard consistency test, setting time test, soundness test, and compressive strength test, Storage of cement and effect of storage on properties of cement. 1.4 Field applications of different types of cements: Rapid hardening cement, Low heat cement, Portland pozzolana cement, sulphate resisting cement, blast furnace slag cement, High alumina cement, White cement.	Lecture Using Chalk-Board Hands-on Site/Industry Visit Collaborative learning Video Demonstrations Case Study

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Identify the type of given aggregate samples based on source, shape and size. TLO 2.2 Explain the methodology to suggest suitability of given fine aggregate. TLO 2.3 Explain the methodology to suggest suitability of given coarse aggregate.	Unit - II Aggregates 2.1 Aggregates: Requirement of good aggregates, Classification according to source, size and shape. 2.2 Fine aggregates: Properties, size, specific gravity, bulk density, water absorption and bulking, fineness modulus and grading zone of sand by sieve analysis, silt content in sand and their specification as per IS 383, bulking of sand. Concept of crushed Sand. 2.3 Coarse aggregates: Properties, size, shape, surface texture, water absorption, soundness, specific gravity, bulk density, fineness modulus of coarse aggregate by sieve analysis, grading of coarse aggregates, crushing value, impact value and abrasion value of coarse aggregates with specification.	Lecture Using Chalk-Board Hands-on Video Demonstrations Site/Industry Visit Collaborative learning Demonstration

CONCRETE TECHNOLOGY**Course Code : 313322**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Justify use of different grades of concrete and their properties for given applications, TLO 3.2 Select w/c ratio for a given grade of concrete, with justification TLO 3.3 Interpret the given data obtained from test on given type of concrete sample. TLO 3.4 Describe the factors affecting overall durability of given type of concrete. TLO 3.5 Explain the given method of concrete mix design for the given situation. TLO 3.6 Describe the need of NDT for the given field situation	Unit - III Concrete and its testing. 3.1 Concrete: Necessity of supervision for concreting operation, different grades of concrete (ordinary Concrete, standard concrete and high strength concrete) as per provisions of IS 456. 3.2 Water cement ratio, Duff Abraham w/c law, significance of w/c ratio, selection of w/c ratio for different grades of concrete and different exposure conditions. 3.3 Properties of fresh concrete: Workability, Factors affecting workability of Concrete. Determination of workability of concrete by slump cone test and compaction factor test. Range values of workability requirement for different types of concrete works. Segregation, bleeding and preventive measures. 3.4 Properties of Hardened concrete: compressive strength, durability, impermeability and dimensional changes of concrete, creep and shrinkage. 3.5 Concrete Mix Design objectives, methods of mix design, study of mix design procedure by I.S. method as per I.S. 10262-(Only procedural steps) 3.6 Testing of concrete: Significance of testing, determination of compressive strength of concrete cubes at different ages, interpretation and co-relation of test results. Non- destructive testing of concrete: Importance of NDT, methods of NDT(only List of methods)	Lecture Using Chalk-Board Video Demonstrations Hands-on Site/Industry Visit Collaborative learning Case Study

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the sequential operations of concreting in given situation. TLO 4.2 Explain the given type of form works and stripping time. TLO 4.3 Identify the type of construction joint to be used in given situations of concreting works, with justification.	Unit - IV Quality Control of Concrete 4.1 Concrete operations: Batching – Definition and types of Batching, Mixing – Types of Mixing and Mixers, Transportation- Modes of Transportation of concrete, precautions to be taken during transportation, Placing- placing of concrete in form work, precautions to be taken while placing of concrete, Compaction of concrete- methods of compaction, care to be taken during compaction, Finishing of concrete- purpose of finishing, types of Finishing, Curing of concrete- definition of curing, necessity of curing, different methods of curing and their application 4.2 Formwork for concreting: Different types of formworks for beams, slabs, columns. Materials used for form work. Requirement of good form work. Stripping time for removal of formworks per IS 456-2000 provision for different structural members. 4.3 Joints in concrete construction: Types of joints, joining old and new concrete, methods of joining, materials used for filling joints.	Lecture Using Chalk-Board Video Demonstrations Hands-on Site/Industry Visit Collaborative learning Cooperative Learning

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Justify the need of given admixtures in concrete mix. TLO 5.2 Describe the characteristics and uses of given type of Mineral admixtures in concrete. . TLO 5.3 Select the type of Special Concrete in the given situation with justification.	Unit - V Admixture in concrete and Special Concrete 5.1 Admixture in concrete: Purpose, properties and application for different types of admixture such as accelerating admixtures, retarding admixtures, water reducing admixture, air entraining admixture, plasticizers and super plasticizers, (IKS*: Paste of Vegetable Extract and Jaggery used as a admixture in the construction of Vadakumnathan temple, Tirussur, Kerala, India .) 5.2 Cementitious (Mineral) Admixtures: Fly ash, Silica Fume, Blast furnace slag, its use, properties and effects 5.3 Special Concrete: Properties, advantages and limitation of the following types of Special concrete: Ready mix Concrete, Fiber Reinforced Concrete, High performance Concrete and self-compacting concrete, light weight concrete.	Lecture Using Chalk-Board Video Demonstrations Case Study Site/Industry Visit Collaborative learning Presentations

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Check suitability of cement based on its fineness.	1	Determine fineness of cement by Blaine's air permeability apparatus or by sieving.	2	CO1
LLO 2.1 Propose the water cement ratio for the given type of cement.	2	*Determine standard consistency, initial and final setting times of OPC.	2	CO1

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Undertake the testing on the given type of cement to determine its compressive strength.	3	*Determine compressive strength of ordinary Portland cement.	2	CO1
LLO 4.1 Measure the volumetric changes in the given sample of cement.	4	Determine the soundness of ordinary Portland cement (OPC)	2	CO1
LLO 5.1 verify the suitability of given sample of sand to prepare concrete	5	Determine silt content of given sample of sand by volume.	2	CO2
LLO 6.1 Undertake the required test to estimate the quantity of sand for preparing concrete	6	Determine bulking of the given sample of sand.	2	CO2
LLO 7.1 Use relevant aggregate for concrete mix design.	7	Determine bulk density of fine and coarse aggregates.	2	CO2
LLO 8.1 Decide percentage of absorbed water in aggregate and able to decide Water cement ratio.	8	Determine water absorption of fine and coarse aggregates.	2	CO2
LLO 9.1 Decide Grading of fine aggregates for quality of concrete Mix.	9	*Determine Fineness modulus of fine aggregate by sieve analysis.	2	CO2
LLO 10.1 Decide Grading of coarse aggregates for quality of concrete Mix.	10	*Determine Fineness modulus of coarse aggregate by sieve analysis.	2	CO2
LLO 11.1 Decide the suitability of coarse aggregate for wearing or non-wearing surfaces.	11	*Determine aggregate impact value or Crushing Value or Abrasion value	2	CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 12.1 Decide the suitability of coarse aggregate for concrete work based on its size and shape.	12	*Determine aggregate elongation index and flakiness index.	2	CO2
LLO 13.1 Decide suitability of concrete according its workability in different situations.	13	*Determine workability of concrete by slump cone test or Compaction factor test.	2	CO3
LLO 14.1 Measure the ability of concrete to resist the compressive loads.	14	*Determine compressive strength of concrete for 7 days	2	CO3
LLO 15.1 know the components parts of RMC and its functioning.	15	*Field Visit/video demonstration on RMC plant to understand the components and its functioning.	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

specific learning / Skills Development (Self Learning)

- Prepare a list of IS codes with their Specifications required for Cement, Aggregate and concrete (Any three codes for each).
- Study of Sieve Analysis for WMM (Wet Mix Macadam) and GSB(Granular sub base) courses.

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- Study of water quality parameters used in concrete as per IS 456.
- Study of various methods of NDT such as Rebound Hammer method and Ultrasonic pulse velocity test.
- Prepare a study report on hot & cold weather concrete.

Micro project

- Market survey to select type of cement for various types of construction works.
 - Visit to site under construction to observe concreting operations.
 - Visit to site under construction to observe the quality of fresh concrete.
 - Visit to site under construction to observe form work, scaffolding used and joints in concrete.
 - Search the software/freeware for the course content and prepare report stating their applications.
 - Visit to site under construction to collect detail information about the ingredients of concrete mix.
 - Market survey to select types of admixture for relevant construction work.
 - Prepare mix design for the grade given by course teacher.
 - Visit to stone crusher to observe its manufacturing.
-

CONCRETE TECHNOLOGY**Course Code : 313322****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Blaine's air permeability apparatus as per IS: 4031 (part 1)-1999, and sieve no. IS 90 micron - IS Brass Sieve (200 mm dia), 90 Micron size.	1
2	I S sieve set (sizes- 80 mm, 40 mm, 20 mm, 10 mm, 4.75 mm, 2.36 mm, 1.18 mm, 600 μ , 300 μ , 150 μ , 90 μ and pan) , sieve shaker with adaptors	1,9,10,11
3	Aggregate impact testing apparatus with mould,	11
4	Aggregate crushing mould	11
5	Los Angeles abrasion testing machine	11
6	Elongation gauge and thickness gauge.	12
7	Slump cone(top dia.100mm, bottom dia.200mm, Height 300mm)	13

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Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
8	Compaction factor test apparatus	13
9	Table vibrator, moulds(150mm x150mmx 150mm)	14
10	Vicats apparatus- VICAT mould of dia. 80 mm & 40 mm high glass base plate, initial needle, final needle. Consistency plunger M.S. base plate (non porous) of weight 300 gm. Vicat mould split type with camping ring.	2
11	Measuring Cylinder 1000 ml capacity	2,3,5,6
12	Measuring Cylinder of 100 ml capacity	2,3,5,6
13	Compression testing machine-2000 kN capacity, Cement mortar cube vibrator-, moulds size 50 cm ² (7.07 cm x 7.07 cm)	3,14
14	Curing Tank	3,14
15	Le chatelier apparatus	4
16	Hot Air Oven	5,6,7,8,9,10,11,12
17	Density basket as per IS specification	7
18	Weighing Balance	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Cement	CO1	6	2	4	4	10
2	II	Aggregates	CO2	8	2	4	6	12
3	III	Concrete and its testing.	CO3	16	4	8	12	24
4	IV	Quality Control of Concrete	CO4	8	2	4	6	12

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Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
5	V	Admixture in concrete and Special Concrete	CO5	7	4	8	0	12
Grand Total				45	14	28	28	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Term work, Assignment, Micro project, SLH (60% Weightage to process and 40% weightage to product), Question and Answer

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test), Practical Exam,

XI. SUGGESTED COS - POS MATRIX FORM

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Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	2	2	2		2			
CO2	3	2	2	2	2		1			
CO3	2	2	3	2	2	1	2			
CO4	2	2	2	2	2	2	2			
CO5	2	2	2	1	2	1	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Gambhir, M.L.	Concrete Technology	Tata McGraw Hill Publishing Co. Ltd., New Delhi, ISBN-13: 978-1-259-06255-1
2	Shetty, M.S	Concrete Technology	S. Chand and Co. Pvt. Ltd., Ram Nagar, New Delhi-110055 ISBN, : 978-8-121-90003-4

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Sr.No	Author	Title	Publisher with ISBN Number
3	Santhakumar ,A. R	Concrete Technology	Oxford University Press, New Delhi ISBN-13: 978-0-195-67153-7
4	Neville, A. M. and Brooks, J.J.	Concrete Technology	Pearson Education Pvt. Ltd., New Delhi ISBN 978-0-273-73219-8
5	Job Thomas	Concrete Technology	CENAGE Publication ISBN-13:978-81-315-2668-2

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://archive.nptel.ac.in/courses/105/102/105102012/	NPTEL Concrete Technology course by Dr. B. Bhattacharjee, Department of Civil Engineering, IIT Delhi
2	https://youtu.be/sl0smPfvVAo?list=PLkyVnO47pDX9YJglk1o2iYzWgABo5I_xA	Testing of Cement by NITTTR Chandigarh
3	https://www.nitttrchd.ac.in/sitenew1/nctel/civil.php	Practical's on Cement/Concrete/Aggregate by NITTTR Chandigarh
4	https://www.youtube.com/watch?v=yzpWGrh9j6Y	Workability of Concrete – Slump Cone Test. By Ultra tech cement.
5	https://www.youtube.com/watch?v=M9hkvS_OLmk	Ultrasonic Pulse Velocity Test on Concrete.
6	https://youtu.be/6iThtydES5c?si=OAOv5EW3lukFiFBD	Fineness of Cement as per IS 4031-Part1 by Sieving method
7	https://cs-iitd.vlabs.ac.in/List%20of%20experiments.html	Virtual laboratory practical on concrete technology

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Sr.No	Link / Portal	Description
8	https://www.researchgate.net/publication/283109018	Research Article on Knowing from the past – Ingredients and technology of ancient mortar used in Vadakumnathan temple, Tirussur, Kerala, India (for IKS) in Journal of Building Engineering, ELSEVIER Publication
9	https://youtu.be/83AcFYK-Eno?si=5HUfRSWIQOLftjvc	Rebound Hammer Test
10	https://youtu.be/KHDdhYVP_GI?si=p3ADC6IkclMTKnai	Visit to Ready mix concrete (RMC) batching plant How RMC plant works?
11	https://youtu.be/SR_Eq9Z7tbM?si=FQlrkUkEBx5zidt8	Visit to RMC Concrete Batching Plant Animation video
12	https://youtu.be/rdtDV-bE0wo	The Overview of the Stages involved in Manufacture of Concrete
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

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HIGHWAY ENGINEERING**Course Code : 313323**

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Third
Course Title	: HIGHWAY ENGINEERING
Course Code	: 313323

I. RATIONALE

Efficient network of road is the life line of any nation. Highway engineering is one of the most favored-effective and economical modes of land transportation. It is instrumental in determining the economic development of the country. Highway and Transport facilities are an important part of urban infrastructure. Highway engineering is one of the branches of Civil engineering which deals with the process of design, construction, and maintenance of different types of roads. The basic requirements of efficient transportation are speed, safety, and comfort. It provides door to door service and connects to inaccessible parts of the country. Hence it is essential for Civil engineering student to acquire the knowledge of this course.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Undertake the construction of the given type of pavements including its maintenance (Roads).

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Identify the roads based on recommendations of IRC.
- CO2 - Implement geometrical features of different Highways.
- CO3 - Observe the various road construction activities.

HIGHWAY ENGINEERING**Course Code : 313323**

- CO4 - Suggest the traffic control devices and intersections based on traffic flow survey data.
- CO5 - Suggest the relevant precautionary measures to control the drainage based on inspection to maintain the given section of roads.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total	FA-PR		SA-PR		SLA					
												Max	Min	Max	Min	Max	Min	Max	Min		
313323	HIGHWAY ENGINEERING	HEN	DSC	3	-	2	1	6	3	3	30	70	100	40	25	10	-	-	25	10	150

HIGHWAY ENGINEERING**Course Code : 313323****Total IKS Hrs for Sem. : 1 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the relevant modes of transportation with its importance. TLO 1.2 Explain Role of IRC, NHAI in the development of roads in India. TLO 1.3 Classify the Roads TLO 1.4 Decide the relevant factors influencing the ideal alignment of road pavement.	Unit - I Introduction to Highway Engineering 1.1 Different modes of transportation and importance of road transportation. 1.2 History of the Road development in India and the agencies involved in this work. (*IKS- Ancient Roads) 1.3 General classification of Roads. 1.4 Alignment: Definition, Requirements, importance etc. of an ideal road alignment and the factors affecting road alignment.	Video Demonstrations Lecture Using Chalk-Board Site/Industry Visit Presentations Cooperative Learning
2	TLO 2.1 Illustrate various functional terms related to geometrics of the given type of highway with sketches. TLO 2.2 Sketch /Draw the cross sections of roads in embankment and cutting for the given site condition.	Unit - II Geometric Elements of Highway 2.1 Definition, purpose, types, and its IRC recommendation of Various geometric elements /Technical terms of road pavement :Permanent way/right of way, road formation, Carriageway width, Road margin, side slopes, side gutter, Camber, Gradient, Super elevation, Curves, Road Widening, Sight distance, Design speed and factors affecting design speed 2.2 Standards cross-sections of EXPRESSWAY, NH, SH, MDR, ODR and VR in embankment and cutting.	Video Demonstrations Presentations Lecture Using Chalk-Board Site/Industry Visit Collaborative learning Case Study

HIGHWAY ENGINEERING**Course Code : 313323**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Undertake the specific test on given sample of bitumen to check its suitability in road construction. TLO 3.2 Differentiate between the flexible and rigid pavement. TLO 3.3 Explain the importance of each layer in given type of road. TLO 3.4 Explain the construction of the given pavement including joints and sealers. TLO 3.5 Describe the components of the hill road in given situation. TLO 3.6 Suggest relevant measures to control the landslide in the given situation.	Unit - III Construction of Road Pavements 3.1 Overview of highway construction materials: Aggregate, Cement, Sand, Water, Soil, Bitumen: Ductility, Flash and Fire Point Test, Softening Point Test, Extraction Test, Grade of Bitumen, Emulsion, Cutback, Tar. 3.2 Definition, Types, Structural Components of pavement and their functions. 3.3 Construction of WBM road, Flexible pavement / Bituminous Road. 3.4 Construction of Rigid Pavement (Cement Concrete): methods of construction, Alternate and Continuous Bay Method, Construction joints, filler and sealers. 3.5 Components, functions of Hill roads. 3.6 Landslides: define, Causes, Types, Prevention of landslides.	Video Demonstrations Presentations Lecture Using Chalk-Board Hands-on Site/Industry Visit Case Study

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Conduct traffic volume survey at the given road intersection. TLO 4.2 Justify the importance of PCU in traffic volume study to suggest the solution to the given problem. TLO 4.3 Justify the importance of traffic control devices. TLO 4.4 Justify the Intersections in the given situation.	Unit - IV Traffic Engineering 4.1 Traffic Volume Study (TVS): Definition, Purpose of TVS, method of TVS. 4.2 Passenger Car Unit (PCU) and factors affecting it. 4.3 Traffic control devices and its types: road signs, marking, Signals, Traffic Island. 4.4 Road Intersections: Cloverleaf, Diamond and Trumpet Interchange	Model Demonstration Video Demonstrations Lecture Using Chalk-Board Hands-on Cooperative Learning Case Study

HIGHWAY ENGINEERING**Course Code : 313323**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Describe with sketches the relevant method of providing drainage in the given type of Road. TLO 5.2 Identify the causes of failure based on inspection of given type of pavement. TLO 5.3 Suggest the maintenance and repair works required for given type of defective road.	Unit - V Road Drainage and maintenance 5.1 Drainage-Definition, necessity, types and sketch of drainage system. 5.2 Failure of flexible and Rigid pavement and Its Causes and preventive measures. 5.3 Necessity of Maintenance of Road & its Classification.	Video Demonstrations Presentations Lecture Using Chalk-Board Site/Industry Visit Cooperative Learning Case Study

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Draw the sketches showing standard cross sections of NH, SH, MDR, ODR, VR.	1	*Draw the sketches showing standard cross sections of NH, SH, MDR, ODR, VR in embankment and cutting on A3 size sketch book.	2	CO1 CO2
LLO 2.1 Identify components of Road.	2	*Visit the existing road to identify the component parts of road and prepare photographic report.	2	CO1 CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Interpret the result of Softening point test on bitumen.	3	*Conduct Softening point test on bitumen.	2	CO3
LLO 4.1 Interpret the result of Penetration test on bitumen.	4	*Conduct Penetration test on bitumen.	2	CO3
LLO 5.1 Interpret the result of Flash and Fire Point on bitumen.	5	Conduct Flash and Fire Point test on bitumen.	2	CO3
LLO 6.1 Interpret the result of Ductility test on bitumen.	6	Conduct Ductility test on Bitumen.	2	CO3
LLO 7.1 Interpret the result of Bitumen Extraction Test on bitumen.	7	*Conduct Bitumen Extraction Test.	2	CO3
LLO 8.1 Write the function of each layer of Pavement. LLO 8.2 Observe the working of Highway Construction machineries. LLO 8.3 Onsite testing of Material and sequential method of Construction.	8	*Visit the road under construction to identify layers in the flexible pavement/Rigid Pavement. prepare photographic report consisting of Materials, Machineries used, Method of Construction and on-site testing of Materials.	2	CO1 CO2 CO3
LLO 9.1 Identify the components of Hill Roads.	9	Visit the hill road to study its components, geometrics and prepare the photographic report containing details.	2	CO1 CO2 CO3

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Perform traffic volume survey for a road intersection.	10	*Carry out Traffic Volume Study (minimum two hours of peak period) for an important road intersection or roadway in your city/ town/ village.	2	CO4
LLO 11.1 Analysis traffic volume data and Interpret data.	11	Analysis traffic volume data obtained from above experiment.	2	CO4
LLO 12.1 Examine and relate the meaning of the traffic signs, road markings, islands in your city.	12	*Draw and identify the traffic signs, road markings, islands, intersections in your city/ town/ village and prepare the photographic report.	2	CO4
LLO 13.1 Inspect the existing road drainage system in your area and identify its type.	13	Visit the road of any one type flexible or rigid to know the road drainage system.	2	CO1 CO2 CO5
LLO 14.1 Identify the defects in road.	14	*Visit the road to identify the defects in road and suggest the possible remedial measures for it and prepare photographic report.	2	CO1 CO2 CO5
LLO 15.1 Suggest possible repairs and maintenance of the road.	15	*Suggest possible repairs and maintenance of the road visited in your city/ town/ village.	2	CO2 CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

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VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)**Micro project**

- Develop Highway Intersection Model for smooth traffic flow.
- Differentiate geometric terms of different roads (NH, SH, MDR, ODR, VR) passing through your area.
- Advance Techniques of repairs like CBTR, White topping, Preventive maintenance, etc.
- Identify mode of transportation other than land transportation.
- List the National Highways/State Highways passing through Maharashtra.
- Collect the information relevant to transportation engineering about ongoing and completed road projects (Samrudhi-Mahamarg, Golden Quadrilateral, etc.).
- Role of MSRDC, NHA and IRC in development and construction of roads.
- Asian Highways (AH) analysis.

Assignment

- IS Codes used for Planning, designing, construction and maintenance of roads.
- Illustrate the terms 1) Granular Sub Base-GSB 2) Wet Mix Macadam- WMM.
- New techniques and machineries used for rapid Highway Maintenance around the world.
- Evaluate the camber and gradient of any one road of each type of pavement in your area of college.
- Develop the photographic model of typical pavement structure for actual visited site.
- Make a list of Mega Highway Projects around the world and advanced techniques/machineries used in it.
- Participate in RTO Safety week and Study its activities.
- Observe construction techniques of WBM/Flexible/ Rigid Road.

HIGHWAY ENGINEERING**Course Code : 313323****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Ring and Ball test apparatus (Hot plate 160mm dia. with magnetic stirrer, brass ring, steel ball and glass vessel 600ml and glass thermometer +800c.	3
2	Digital Analytical Weighing Scale, Accuracy: 1 Mg capacity : 30 g to 300 g	3,4,5,6,7
3	1) Lab Safety products (Heat Resistant Gloves, PPE Kit for Laboratory work, etc.) 2) Laboratory items-Lab Utensils and Hand Tools and Sample Containers and Bags	3,4,5,6,7
4	Standard Penetrometer with penetration needle 100gm weight, container 55mm dia. and 53mm ht. as per IS:1203.	4
5	Pensky Marten's Flash and Fire Point test apparatus 100x200x240mm with measurement range 0-95 as per IS:1209-1953	5

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Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
6	Ductility Testing Machine with ductility mould and base plate	6
7	Bitumen Extraction Test Apparatus: - Centrifuge Extractor, Electrical Operation, Capacity 1500g, with a Dimmer stat for speed control from 2,400 to 3,600 rpm. Suitable for operation on 230 V, 50 Hz, Single Phase, A.C. supply.	7

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Introduction to Highway Engineering	CO1	3	2	4	0	6
2	II	Geometric Elements of Highway	CO2	15	6	8	8	22
3	III	Construction of Road Pavements	CO3	15	6	8	8	22
4	IV	Traffic Engineering	CO4	6	2	4	4	10
5	V	Road Drainage and maintenance	CO5	6	2	4	4	10
Grand Total				45	18	28	24	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Termwork, Assignment, Microproject (60% Weightage to process and 40% weightage to product), Question and Answer.

Summative Assessment (Assessment of Learning)

- Pen and Paper Test (Written Test)

HIGHWAY ENGINEERING**Course Code : 313323****XI. SUGGESTED COS - POS MATRIX FORM**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	-	-	-	2	-	3			
CO2	3	-	1	2	1	-	3			
CO3	3	3	2	3	2	1	3			
CO4	3	3	3	2	3	3	3			
CO5	3	3	3	3	3	3	3			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Khanna S.K., Justo, C E G and Veeraragavan, A.	Highway Engineering	New Chand and Brothers, Rookie, 2010, ISBN 978-8185240800

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Sr.No	Author	Title	Publisher with ISBN Number
2	Kadiyali, L.R.	Traffic Engineering and Transport Planning	Khanna Publishers, New Delhi, 2008, ISBN: 978-8174092205
3	Duggal, Ajay K. and Puri, V. P.	Laboratory Manual in Highway Engineering	New Age International (P) Limited, Publishers, New Delhi, 2010, ISBN: 9788122403107
4	Raji A. K. and K. K. Babu	Transportation engineering theory and practice)	AICTE New Delhi ISBN 978-81-960576-1-9
5	N L Arora	Transportation engineering	New India Publishing House, New Delhi ...

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://iksindia.org/index.php	Indian Knowledge Systems (IKS)
2	https://www.youtube.com/watch?v=acfJIG9o8iw	Flakiness and Elongation Index of Aggregate
3	https://www.youtube.com/watch?v=TE8zYxUJHt0 https://ts-nitk.vlabs.ac.in/exp/ductility-test/	Ductility test on Bitumen.
4	https://www.youtube.com/watch?v=-yBXl4z70mI https://ts-nitk.vlabs.ac.in/exp/softening-point-test/	Softening point test on bitumen.
5	https://www.youtube.com/watch?v=9HZE6DNfF5U https://ts-nitk.vlabs.ac.in/exp/penetration-test/	Penetration test on bitumen.
6	https://www.youtube.com/watch?v=PR7q4-ilENA	Flash and Fire Point test on bitumen.
7	https://www.youtube.com/watch?v=JEySduXuxCc&t=563s https://www.youtube.com/watch?v=d48qDaiDyVI	Bitumen Extraction Test
8	https://www.youtube.com/watch?v=2VehMMP70HE&list=PLLy_2iUCG87C7nApYQjgkDA0p67fMaXnE	Geometric Design of Highways By Prof. Rajat Rastogi IIT Roorkee

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Sr.No	Link / Portal	Description
9	https://www.youtube.com/watch?v=5zKC_aq4ypM&list=PLE88643285BC70E0F	Transportation Engineering and Road development Process by IIT Kharagpur
10	https://crridom.gov.in/	CSIR-Central Road Research Institute
11	https://www.irc.nic.in/	Indian Roads Congress (IRC)
12	https://nhai.gov.in/#/	National Highway Authority of India (NHAI)
13	https://msrdc.in/1307/Home	Maharashtra State Road Development Corporation Ltd.
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

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CONSTRUCTION MANAGEMENT

Course Code : 313010

Programme Name/s	: Civil Engineering/ Civil & Rural Engineering/ Construction Technology/ Civil & Environmental Engineering/
Programme Code	: CE/ CR/ CS/ LE
Semester	: Third
Course Title	: CONSTRUCTION MANAGEMENT
Course Code	: 313010

I. RATIONALE

The construction industry is in continuous need of skilled professionals, capable of managing projects efficiently in the capacity of project engineers, construction managers, site supervisors, and project coordinators, among others. It is required for a civil engineer to plan, manage and execute Civil Engineering works with utmost precision within the time frame so as to optimize the resources. Therefore, it is necessary to develop a perfect blend of knowledge, skills, and attitudes in the form of the competencies among the learners to tackle with such engineering projects effectively and efficiently leading towards sustainable development. This course will help in developing these basic competencies among the students which will enable them to get an employment in the market.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Manage the given construction project using the relevant techniques of construction management.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Conduct the project feasibility analysis of the given project.
- CO2 - Apply the relevant scheduling technique in the given situation to decide the ethical element of the project.
- CO3 - Manage the inventory using relevant inventory control techniques.

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- CO4 - Execute the project as per the prevailing safety practices

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
				CL	TL	LL					Practical				Based on SL						
											FA-TH	SA-TH	Total				FA-PR		SA-PR		
													Max	Max	Max	Min	Max	Min	Max	Min	
313010	CONSTRUCTION MANAGEMENT	CMA	DSC	1	-	2	1	4	2	-	-	-	-	-	25	10	-	-	25	10	50

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

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CONSTRUCTION MANAGEMENT**Course Code : 313010****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the term, “Project Life Cycle” with its importance. TLO 1.2 Identify the characteristics of the given project with relevant constraints. TLO 1.3 Select the project from the available options based on feasibility analysis. TLO 1.4 Justify the importance of project management frameworks and standards. TLO 1.5 Select the relevant type of equipment for the given type of activity of project.	Unit - I Project Initiation and its feasibility 1.1 Project: Basic Definitions, management functions, ethics, project life cycle- Project Initiation, Project Planning, Project Executing, Project Monitoring and Controlling, Project Closing. 1.2 Project Characteristics and Constraints- Scope, time, cost, Quality; Stakeholder. 1.3 Project Feasibility Analysis- Market analysis, Financial analysis- Net Present Value(NPV), Payback Period, Examine the business problem/opportunity, Identify the requirements, undertake a feasibility study, Rank the feasibility results- Define the criteria, give ranking scores, Identify the feasibility outcome. 1.4 Project Management Frameworks and Standards, Project Management consultant (PMC)- roles & responsibilities. 1.5 Equipment used in executing the Civil Engineering projects- •Earth moving equipment (Tractor, Bulldozer, Scrapers, Excavators), • Hauling Equipment (Drum trucks, Front end loader, Conveyor belt), • Concreting Equipment (RMC mixer, Concrete pump), • Hoisting Equipment (Lifting & lowering equipments, Cranes).	Lecture Using Chalk-Board Presentations Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Identify the broad activities involved in given construction project. TLO 2.2 Apply the relevant technique of analysis to get the required information about the given project. TLO 2.3 Explain the process of developing the critical path line in solving the given problem.	Unit - II Project Management and Scheduling 2.1 Broad activities in construction work – Earthwork, Foundation, RCC Work, Brick work, Scaffolding, Plastering, Painting etc & duration required for complete the activity 2.2 Methods of Scheduling-Gantt Chart, Bar chart, Development of Bar charts and Gantt chart, Merits & limitations of Bar chart & Gantt chart. 2.3 Concept of CPM & PERT: Introduction to Critical path method (CPM), Program evaluation & review techniques (PERT), Network Diagramming of Projects Activity-on-arrow (AOA) Diagrams- Concept of Activity and Event, Time-Analysis of Networks- Forward Pass, Backward Pass, Probabilistic Durations- Optimistic Time, Pessimistic Time, Most Likely Time, Project Scheduling- ES and LS Schedules as Limits, Resource Scheduling, Time/Cost Trade-off	Lecture Using Chalk-Board Video Demonstrations Hands-on Presentations
3	TLO 3.1 Apply the relevant material management techniques in the given construction project. TLO 3.2 Apply the material management technique for rebar in the given construction project..	Unit - III Material Management 3.1 Material Management-Introduction, Inventory and inventory control, EOQ (Economic order of quantity), ABC technique, V-E-D analysis, Just in Time Strategy (JIT), Store management & various records related to store management 3.2 Rebar Management: Wastage of steel on site, Preventive measures to avoid the wastage of steel. Rebar, Importance of rebar, Fabrication and transportation of Rebar. Understanding rebar drawings.	Lecture Using Chalk-Board Video Demonstrations Presentations Site/Industry Visit

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Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the remedial measures with preventative strategies for the relevant identified cause of accidents on construction sites. TLO 4.2 Follow the relevant legal provisions related to labor laws in project execution.	Unit - IV Safety and labour laws in construction 4.1 Importance of Safety in construction work, causes of accidents on construction site & remedial measures, precautions to avoid accidents at site, safety policies. 4.2 Introduction to Labour laws related to construction- Workman Compensation Act, Minimum Wages Act, The Building and Other Construction Workers' (Regulation of Employment and Conditions of Service) Act, Janshree Vima Yojna	Lecture Using Chalk-Board Video Demonstrations Presentations Site/Industry Visit

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the roles and responsibilities of manpower required for a construction project	1	*Draw the flow chart of manpower required for a given type of project	2	CO1
LLO 2.1 Select the relevant resources required for foundation or RCC or Brickwork for given construction project/data	2	*Draw the resource allocation plan for Foundation or RCC or Brickwork activity on construction site	2	CO1

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 3.1 Select the relevant resources required for Structural members of beam or column for given construction project/data	3	Draw resource allocation plan for Structural members for beam or column or slab for G+5 activity on site	2	CO1
LLO 4.1 Select the relevant resources required for plastering or flooring for given construction project/data	4	Draw resource allocation plan for plastering or painting or flooring activity on site	2	CO1
LLO 5.1 Select different equipments used to carry out the construction of building	5	Write a brief report of site inspection with special reference to construction equipments used in the project with relevant photographs, video etc	2	CO1
LLO 6.1 Apply the concept of bar chart/Gantt chart to get the required information about the given project.	6	Draw the bar chart / Gantt chart for the activities of given construction project by using MS Excel/MS Project	2	CO2
LLO 7.1 Apply the concept of network diagram to get the required information about the given project.	7	Forecast the time duration required for various activities of the given construction project to represent them through a network diagram	2	CO2
LLO 8.1 Determine/Find the duration of the project and the sequence of critical activities.	8	*Solve the numerical on CPM for finding duration of project and Critical path of the any one problem for the given data	2	CO2
LLO 9.1 Apply the concept of PERT technique to get the required information about the given project.	9	*Solve the numerical on PERT to represent the sequence of activities and critical path of the any one problem for the given data.	2	CO2

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Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 10.1 Determine the Economic Order Quantity (EOQ) based on the given data	10	*Determination of EOQ (Economic order quantity) based on the given data. (Solve one Numerical)	2	CO3
LLO 11.1 Identify the most important product in given construction project/data	11	*Carry out the ABC analysis for the given problem/data (Solve one Numerical)	2	CO3
LLO 12.1 Apply the principles of management for rebar procurement on the given site.	12	Develop rebar procurement plan for the given construction site. (Manually or by using any open- source software)	2	CO3
LLO 13.1 Select the causes and remedial measures for given construction project/data	13	*Prepare a report on minimum five expected causes of accidents on construction sites with their remedial measures. (Visit any one Industrial/Residential/Public construction building)	2	CO4
LLO 14.1 Select the action plan measures for given construction project/data	14	*Prepare a brief report of the observation made on site with respect to safety on site (Visit any one Industrial/Residential/Public construction building)	2	CO4
LLO 15.1 Select various safety devices used at given construction site	15	Prepare the charts/report on various safety devices used at given construction site (Visit any one Industrial/Residential/Public construction building)	2	CO4
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

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VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)**Micro project**

- Prepare a report on different forms of inventory storage along with your interpretation.
- Collect the information about latest safety measures adopted at construction project.
- Collect information and prepare a report on any one top construction companies in India.
- Compare any 3 construction management software.
- Use any one free open ware software to collect information about modern techniques of material management like JIT/SAP/ERP/MSP/MSEXCEL/Primavera.
- Collect information and prepare a report on various construction equipments used in construction industry.
- Use any one free open ware software of Construction Management to prepare the scheduling of a project.
- Use any one free open ware software of Construction Management to determine the critical path for the given construction project.
- Interpret the network figures used in given civil engineering projects.

Assignment

- Prepare a brief report on overview of Construction Professional Practice in India
- Solve the numerical on bar chart, CPM and cost optimization for the given data.
- Collect and interpret various store forms from PWD, WRD, MJP.
- Download the labour laws documents from internet and write a brief summary on it.
- Learn material management module from free open ware website.
- Visit to a Site to study the construction technique and use of major construction equipment.
- Student should watch any 3 learning website link given in XIII learning websites & portal and prepare a brief report on it.
- Prepare a brief report on role of construction industry in national development.

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- Compile minimum 10 safety slogans displayed at various sites with sources and write a brief summary on it.
- Prepare project cost analysis for small construction project.
- Collect & interpret bar chart/CPM network for existing construction project.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Free Open ware software-1. Just-In-Time (JIT), 2. System Application & Product in data processing (SAP), 3. Microsoft Project (MSP)4. Microsoft Excel (MS Excel)5. Primavera6. Enterprise Resource Planning (ERP)	2,3

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
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Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Project Initiation and its feasibility	CO1	6	0	0	0	0
2	II	Project Management and Scheduling	CO2	5	0	0	0	0
3	III	Material Management	CO3	2	0	0	0	0
4	IV	Safety and labour laws in construction	CO4	2	0	0	0	0
Grand Total				15	0	0	0	0

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Term work, Self Learning Assessment (Assignment & Microproject). Note: Each Practical will be assessed considering- 60% weightage to process related and 40 % weightage to product related

Summative Assessment (Assessment of Learning)**XI. SUGGESTED COS - POS MATRIX FORM**

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Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	2	1	2	2	2	2			
CO2	3	3	1	2	2	2	2			
CO3	3	3	1	2	1	2	2			
CO4	2	1	2	2	2	2	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	S.C. Sharma, S.V. Deodhar	Construction Engineering and Management	Khanna Book Publishing Co (P) Ltd. ISBN 9789386173980
2	IGNOU	Project Management	Indira Gandhi National Open University(eKumbh-AICTE)
3	K. K. Chitkara	Construction Project Management- Planning, Scheduling & Controlling	McGraw Hill Education ISBN-10 0070680752, ISBN-13 978-0070680753

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Sr.No	Author	Title	Publisher with ISBN Number
4	L. S. Srinath	PERT And CPM Principles And Applications	East-West Press (Pvt.) Ltd. ISBN-10 8185336202 ISBN-13 978-8185336206
5	Jack Gido, Jim Clements, Rose Baker	Successful Project Management	Cengage Publication ISBN: 9781337363853

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://archive.nptel.ac.in/courses/105/103/105103206/	Construction Method and Equipment Management
2	https://www.youtube.com/watch?v=Cx7i2wXB0kA&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=16	Project Scheduling
3	https://www.youtube.com/watch?v=j6VillXT0Vs&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=22	Accidents in Construction Industry
4	https://www.youtube.com/watch?v=EVsi1QamfU0&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=26	Safety Organization and Safety Officer
5	https://www.youtube.com/watch?v=QoXvRBrFWyI&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=25	Implications of Construction Accidents
6	https://archive.nptel.ac.in/courses/105/104/105104161/	Introduction to planning and scheduling, resource levelling and allocation, crashing of networks
7	http://www.cidc.in/	Construction Industry Development Council (CIDC)
8	https://onlinecourses.nptel.ac.in/noc22_ce39/preview	Safety in Construction Industry

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Sr.No	Link / Portal	Description
9	https://www.youtube.com/watch?v=Tm2HhqMu5Jg	PERT and CPM
10	https://www.youtube.com/watch?v=GAGoqqZSP4&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=3	Overview of steps in execution of a project
11	https://www.youtube.com/watch?v=kuCHsNXeNMc&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=5	Resource Management in Construction Projects
12	https://www.youtube.com/watch?v=Bh_LYZh3KH4&list=PLWnoy5z_3BObBvFtBlowxM05D-q0VAWEs&index=21	Introduction to construction safety
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

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